National University of computer and emerging sciences

Database Systems Lab (CL-2005)
Semester: Spring 2026
Section: BCS-4E
Course Instructor: Mr. Abdullah Shaikh

LAB # 10

Comprehensive Study of PL/SQL: Blocks, Data Types, Conditional Logic, Loops, Procedures, and Functions

Submission Instructions:

- Perform all the assigned tasks carefully.
- Prepare your solutions in a Microsoft Word document, clearly writing the solution of each problem.
- Attach relevant screenshots for every task as proof of execution.
- Ensure that your document is properly formatted (clear headings, proper spacing, and alignment).
- After completing the Word document, convert it into a PDF file.
- Upload the PDF file on Google Classroom (GCR) within the given deadline.
- No submission will be accepted after the deadline.
- Before uploading, rename your PDF file using the following format:
  DB Systems Lab 10 – Your Name – Your Roll Number
  Example: DB Systems Lab 10 – Abdullah Shaikh – 25K-7826

Lab # 10
Comprehensive Study of PL/SQL: Blocks, Data Types, Conditional Logic, Loops, Procedures, and Functions

1. Write a PL/SQL block that accepts an employee_id and displays the employee's salary category along with their department classification based on department_id, and prints a combined status report for that employee.
2. Write a PL/SQL block that processes all employees and displays whether each employee is eligible for technical recognition or standard classification based on job role and salary conditions.
3. Write a PL/SQL block that generates a report for all employees in which each employee is categorized by salary level and assigned a job-based performance label, both shown together.
4. Write a PL/SQL block that evaluates employees and displays a performance status based on salary and commission values, including handling cases where commission is missing.
5. Write a PL/SQL block that calculates total and average salary for each department and displays a budget classification for each department based on the calculated values.
6. Write a PL/SQL block that takes a department_id and generates a detailed employee report showing job classification, salary-based eligibility status, and promotion consideration status.
7. Write a PL/SQL block that produces a full employee analysis report including salary grading, job role classification, and experience-based categorization for all employees.
8. Write a PL/SQL block that calculates and displays bonus eligibility for all employees based on department, salary level, and commission contribution.
9. Write a PL/SQL block that identifies employees whose salary is below company average and displays their job classification along with a priority status based on department.
10. Write a PL/SQL block that generates an employee ranking report showing salary tier, job role category, and departmental classification for each employee.
11. Write a PL/SQL block that produces a hierarchical report of departments and employees, showing salary status and job role classification under each department.
12. Write a PL/SQL block that validates an employee_id and displays full employee status including salary grade, job classification, and promotion eligibility.
13. Write a PL/SQL block that adjusts employee salaries based on department and tenure, and displays the updated salary along with classification details.
14. Write a PL/SQL block that generates an employee experience report classifying employees into junior, mid-level, and senior categories along with their job roles and salary grades.

15. Write a PL/SQL block that produces a consolidated report combining employee status, department classification, job role mapping, salary grading, and bonus eligibility for all employees.
16. Create a stored procedure that accepts an employee_id and displays the employee's full details including first name, job_id, salary, department_id, and hire_date from the HR schema.
17. Create a stored procedure that takes an employee_id and increases the employee's salary by a given percentage, then displays the updated salary along with the employee's name.
18. Create a stored procedure that accepts a department_id and displays all employees belonging to that department along with their job_id and salary.
19. Create a stored procedure that calculates and displays the total number of employees, total salary, and average salary for a given department_id.
20. Create a stored procedure that accepts a job_id and lists all employees having that job role along with their department name, salary, and hire_date.

THE END!